Sahaj Bhadja

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Summary

Software engineer with **2+ years of experience** building **consumer-facing full-stack products**, **real-time workflows**, and **distributed cloud services** across **React, TypeScript, AWS, REST**, and performance-focused backend systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Architected a **multi-tenant AI platform** on Azure SQL and Blob Storage for **500+ users**, reducing dashboard latency from **1.2s to 350ms** through query tuning, indexing, and application-performance optimization.
- Built **FastAPI** integration workflows across PostgreSQL, SQL Server, and cloud storage, processing **5K+ monthly jobs** with validation, debugging, and release checks that sustained **99%+ production reliability**.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Developed and tested a **Flask + React** grading platform serving **150+ instructors**, handling **10K+ submissions** with secure ingestion, user-facing dashboards, and production backend workflows.
- Designed concurrent backend services on **AWS RDS** with REST APIs and **GitHub Actions CI/CD**, sustaining **10K+ requests/day** while reducing turnaround time by **60%**.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform microservice software with **Jenkins**, PyTest, and Mockito, automating build, test, and release flows across **3 operating systems** with **20+ unit/integration suites**.
- Optimized API and ETL workflows using **AWS Glue**, **S3**, and OOP patterns, improving response time by **30%**, boosting pipeline efficiency by **68%**, and reducing technical debt by **20%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a **MERN** resource-management platform with **REST APIs**, WebSockets, and microservices, supporting **8+ business workflows** and synchronizing **2M+ records** across locations.
- Automated Dockerized **Jenkins** testing and deployments for Azure SQL-backed services, improving release reliability while reducing resource wastage by **42%** and increasing lead conversion by **15%**.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built a cross-platform booking platform with **React**, TypeScript, **Flask**, and Firebase, supporting **50K+ booking records** with real-time sync and sub-**200ms** API latency for reservation flows.
- Integrated webhook-driven payment and status updates with Firebase-backed real-time data flows, improving booking confirmation speed by **40%** and reducing manual refund handling by **70%**.

Skills

Languages: Python, Java, C++, JavaScript/TypeScript, SQL, Golang
Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs, WebSockets, Microservices
Frontend: React.js, Next.js, TailwindCSS
Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, SQL Server, Azure SQL
Cloud & Infrastructure: AWS (ECS, Lambda, SQS, Step Functions, RDS, S3), Microsoft Azure, Docker, Kubernetes, GitHub Actions, Jenkins
AI/ML: PyTorch, Transformers, NLP, LLM Systems, Feature Engineering
Engineering Practices: CI/CD, Unit & Integration Testing, Distributed Systems, Performance Optimization, ETL Pipelines, Technical Documentation, Agile

Projects

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB, WebSockets*
- Built a consumer-facing fintech platform supporting **50K+ users** and **1M+ transactions** with real-time updates, WebSocket-driven synchronization, and cloud-native deployment on **Kubernetes**.

HMIS Live | *MERN, TypeScript, Twilio, Sentry*
- Built a user-facing healthcare platform supporting **100+ daily appointment workflows**, integrating ABDM APIs, Twilio alerts, and monitoring while reducing patient record retrieval time by **2x**.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL*
- Built a collaborative document platform over **50K+ documents** with sub-second retrieval, real-time synchronization, and cloud indexing pipelines that cut review time by **60%**.